Silverio Martínez Andrés Grupo 35 Tarea 29

Nombre:

Tema:

Día Mes Año Folio

Tarea 29

1.- Calcular la pendiente de la tangente a la curva de ecuación $y = x^2 - 3$ en el punto $P(1, -2)$

$y = 2x$

$m = y' \Big|_{P(1,-2)} \; 2(1) = 2$

2.- Obtener la ecuación de la recta tangente y de la normal, así como las longitudes de la tangente, de la normal, de la subtangente y de la subnormal a la curva $y = (x-2)^2 + 1$ en el punto $P(3, 2)$ ($x_0$, $y_0$)

$Ec = tg$

$y - y_0 = f'(x_0)(x - x_0)$

Ec. normal $m = 2$

$y - y_0 = -\frac{1}{f'(x_0)}(x - x_0)$

$y' = 2(x-1)'(1)$

$y' = 2x - 2$

$y' \Big|_{P(3,2)} = 2(3) - 4 = 2$

Ec. tg

$y - 2 = 2(x - 3)$

$y - 2 = 2x - 6$

$2x - 6 - y + 2 = 0$

$2x - y - 4 = 0$ → Ecuacion tg

Ec normal

$y - 2 = \frac{-1}{2}(x - 3)$

$2y - 4 = -x + 3$

$2y - 4 + x - 3 = 0$

$x + 2y - 7 = 0$ → Ecuacion normal.

$3T = \frac{2}{2} = 1$

Escaneado con CamScanner

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Tema:

$\overline{PT} = \sqrt{2^2+1^2} = \sqrt{4+1} = \sqrt{5}$

$\overline{SN} = 2(2) = 4$

$\overline{PN} = \sqrt{4^2+2^2} = \sqrt{16+4} = \sqrt{20}$

$\overline{ST} = \frac{y_0}{m}$

$\overline{PT} = \sqrt{y_0^2 + (\overline{ST})^2}$

$\overline{PN} = \sqrt{(\overline{SN})^2 + 4y^2}$

$\overline{SN} = my_0$

3.- Determinar el angulo que forman al cortarse las parabolas

$y^2 = \frac{x}{2}$  $y = \frac{x^2}{4}$

$y = \pm\sqrt{\frac{x}{2}} \to y^2 = \frac{x^4}{16}$

| x | $y = \pm\sqrt{\frac{x}{2}}$ |
|---|---|
| 0 | 0 |
| 2 | $\pm 1$ |
| 8 | $\pm 2$ |

| x | $y = \frac{x^2}{4}$ |
|---|---|
| -2 | 1 |
| -1 | 1/4 |
| 0 | 0 |
| 1 | 1/4 |
| 2 | 1 |

$\pm\sqrt{\frac{x}{2}} = \frac{x^2}{4}$

$\frac{x}{2} = \frac{x^4}{16}$

$16x = 2x^4$

$2x^4 - 16x = 0$

$x(2x^3 - 16) = 0$

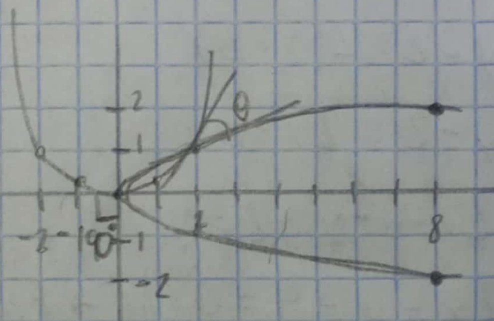
$2x^3 - 16 = 0$

$2x^3 = 16$

$x^3 = 8$

$\boxed{x = 2}$  $\boxed{x = 0}$

$\boxed{y = 1}$  $\boxed{y = 0}$



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$y^2 = \frac{x}{2} \rightarrow y = \pm\sqrt{\frac{x}{2}}$

$y = \frac{x^2}{4} \rightarrow y^2 = \frac{x^4}{16}$

$2yy' = \frac{1}{2}$

$y' = \frac{2x}{4}$

$y' = \frac{1}{4y}$

$y' = \frac{x}{2}$

$m_1 = y'\Big|_{y=1} = \frac{1}{4}$ $m_2 = y'\Big|_{x=2} = 1$

$\theta = \text{ang tg } m_2 - \text{ang tg } m_1$

$\theta = \text{ang tg } 1 - \text{ang tg } (1/4)$

$\theta = 45^\circ - 14^\circ$

$\boxed{\theta = 31^\circ}$

$\theta = \text{ang tg } \frac{m_2 - m_1}{1 + m_2 m_1}$

$= \text{ang tg } \frac{1 - 1/4}{1 + 1(1/4)} = tg^{-1}\left(\frac{\frac{3}{4}}{\frac{5}{4}}\right)$

$tg^{-1} = \left(\frac{3}{5}\right)$

$= 31^\circ$